1D PRD source function — as built

What changed, the bug it exposed, and how to run it

Implementation record · 2026-09-08 · not yet committed / not yet run on the full cube

Code changes

`calc_op_em.py`

- `load_s1d_table(opem_path, wave)` — *new*. Builds the table {`wave[nm], z[km asc], S[Nλ,Nz]`} as $S = \eta/\chi$ directly from `disk_center_test_opem.fits` (single total source function, no line/continuum split), interpolated onto the synthesis grid.
- `calc_op_em(..., s1d=None, dz_cal=0.0)` — pulls `'Height'` from the ray (masked/refined in lockstep) and, when a table is given, overrides $\eta = \chi \cdot S(\lambda', z')$ via `RegularGridInterpolator`. Lookup is velocity-shifted $\lambda' = \lambda - (v/c)\lambda_0$ (same shift as the Voigt profile) and calibration-shifted $z' = h - \Delta z_{\rm cal}$, clamped to the table range (nearest-edge above the FALC top, 2342 km).

`mpi_synt_cube_simple.py`

- `create_ray` now carries `'Height': z` (the curved-ray geometric height, already computed).
- `synth(..., s1d, dz_cal, use_source_function=True)` — real formal solution with the 1D S; legacy $1-e^{-\tau}$ kept behind the flag (`s1d=None`).
- `worker_work(rank, s1d, dz_cal)` threads the table through.
- `__main__` loads the table **once on rank 0**, `comm.bcast`s it to all workers (no worker opens the file), and reads $\Delta z_{\rm cal}$ as an optional **4th CLI argument** (km, default 0).

Latent bug found & fixed: `simple_formal_solution`

Turning on the source function exposed an error in the formal solver: it formed $(1 - e^{-\tau_{\rm cumulative}})\,S$ and multiplied by the cumulative transmission *again* — wrong for optically-thick cells, yielding $I > \max(S)$, which is impossible with no incident radiation. It never mattered before because the old `synth` *discarded* this output and returned $1 - e^{-\tau}$. Replaced with the correct, `expm1`-stable

$$I_\lambda = \sum_i S_i \left(1 - e^{-\Delta\tau_i}\right) e^{-\tau_{{\rm upwind},i}}, \qquad \tau_{{\rm upwind},i} = \tau_i - \Delta\tau_i.$$

Now bounded by $\max(S)$. The legacy $1 - e^{-\tau}$ path is unaffected (same τ).

Validation (synthetic columns)

- I finite, ≥ 0, and **bounded by** $\max(S)$ along each ray.
- Core intensity **decays with tangent height**: 4.5×10^{-9} (200 km) $\rightarrow$ 1.6×10^{-9} (1200 km) $\rightarrow$ 1.2×10^{-11} (3000 km) — the physical off-limb falloff, versus the legacy profile saturating at $\sim$1.
- The velocity/height lookup reproduces a direct table interpolation at the shifted (λ, h); $\Delta z_{\rm cal}$ shifts the result as expected.

How to run

```
module load mpi/openmpi-401_gcc-485
mpirun -n <N> python mpi_synt_cube_simple.py <cube.h5> <end> <filename> <dz_cal_km>
```

The 4th argument is optional; omit it for $\Delta z_{\rm cal} = 0$. (A plain shell without the MPI module cannot even import the driver — `mpi4py` needs `libmpi`.)

Open items

- **Calibrate** $\Delta z_{\rm cal}$: align MURaM's $z = 0$ to FALC's by matching the disk-centre profile or the limb-intensity inflection.
- **Performance**: the per-ray 2D interpolation is the hot spot at scale; optimise later, or drop the velocity shift for a rest-frame first run.
- **Not yet run** on `/dat/milic/SUSI_modeling/extended_off_limb_ssd_cube_paper_angle_80.h5`; changes are uncommitted (`calc_op_em.py`, `mpi_synt_cube_simple.py`).